

8)

```
static public int pIter (int x, int n) {
```

```
    int p;      C1
```

```
    if (n == 0)
```

```
        p = 1;  C2
```

```
    else {
```

```
        if (n == 1)
```

```
            p = x; C3
```

```
        else {
```

```
            p = x; C3
```

```
            for (int i = 2; i <= n; i++) {
```

```
                p *= x; C4
```

```
            }
```

```
        }
```

```
    }
```

```
    return p; C5
```

```
}
```

$$T(n) = \begin{cases} C_1 + C_2 + C_5 & n = 0 \\ C_1 + C_3 + C_5 & n = 1 \\ C_1 + \sum_{k=1}^{n-1} C_4 + C_3 + C_5 & n > 1 \end{cases}$$

$$C_1 + C_3 + \sum_{k=1}^{n-1} C_4 + C_5$$

Iteración 1 $i = 2$ $k+1 = n-1$ Iteración 2 $i = 3$ $k = n-1$ Iteración 3 $i = 4$ $k = n-2$ Iteración 4 $i = 5$ Iteración k $i = k+1$

$$C_1 + C_3 + \sum_{k=1}^{n-1} C_4 + C_5 = C_1 + C_3 + (n-1) \cdot C_4 + C_5 =$$

$$= C_4 n + C_4 + C_1 + C_3 + C_5 \quad T(n) = O(n)$$

~~$$T(n) = C_1 + C_3 + \sum_{k=1}^{n-1} C_4 + C_5$$~~

$$T(n) = \begin{cases} C_1 + C_4 + C_5 & n=0 & O(1) \\ C_1 + C_3 + C_5 & n=1 & O(1) \\ C_1 + C_3 + \sum_{k=1}^{n-1} C_4 + C_5 & n \geq 1 & O(n) \end{cases}$$

8)

```
static public int pRec (int x, int n) {
```

```
    if (n == 0)
```

```
        return 1; C1
```

```
    else {
```

```
        if (n == 1)
```

```
            return x; C2
```

```
        else {
```

```
            if ((n % 2) == 0)
```

```
                return pRec (x * x, n/2); C3
```

```
            else
```

```
                return pRec (x * x, n/2) * x; C3 > C4
```

```
        }
```

```
    }
```

```
}
```


$$T(n) = \begin{cases} C_1 & n = 0 \\ C_2 & n = 1 \\ T\left(\frac{n}{2}\right) + C_3 & n > 1 \vee n \bmod 2 = 0 \\ T\left(\frac{n}{2}\right) + C_3 + C_4 & n > 1 \vee n \bmod 2 = 1 \end{cases} \quad \begin{matrix} O(1) \\ O(1) \\ O(\log(n)) \\ O(\log(n)) \end{matrix}$$

Paso 1

$$T\left(\frac{n}{2}\right) + C_3$$

Paso 2

$$1 \left[T\left(\frac{n}{2}\right) + C_3 \right] + C_3 = T\left(\frac{n}{4}\right) + 2C_3$$

Paso 3

$$1 \left[T\left(\frac{n}{4}\right) + C_3 \right] + 2C_3 = T\left(\frac{n}{8}\right) + 3C_3 = T\left(\frac{n}{2^3}\right) + 3C_3$$

Paso i

$$T\left(\frac{n}{2^i}\right) + i \cdot C_3$$

Caso base

$$\frac{n}{2^i} = 1 \rightarrow n = 2^i = \log_2(n) = i$$

Reemplazo

$$T\left(\frac{n}{2^{\log_2(n)}}\right) + \log_2(n) \cdot C_3 = T\left(\frac{n}{n}\right) + \log_2(n) \cdot C_3 =$$

$$= T(1) + C_3 \cdot \log_2(n) = C_2 + C_3 \cdot \log_2(n)$$

$$T(n) = O(\log(n))$$